

Guide to writing a lab report

Modern Physics (PHY 223)

Franklin & Marshall College

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1 Overview

This document is intended to be used as a reference while you write your lab report. We recommend you use the pre-writing (Sec. 2) and post-writing (Sec. 5) checklists to improve the process of writing the report and the quality of your manuscript. In Sec. 3, we propose an order to writing the various sections of your manuscript. Before writing any parts of your report, review the relevant instructions in Sec. 4 as linked in the Table of Contents on the previous page. More information about the appropriate styles for figures and tables are given in Sec. 6.

2 Checklist: Before you write

- ☐ **Review the literature.** You should research your topic thoroughly. This can be done as you are taking and analyzing data.
- ☐ Finish taking your **data**.
- ☐ **Create an initial set of figures** for your report.
- ☐ **Complete your analysis** of the results. This should include full consideration of uncertainty.
- ☐ Based on your results, **write down your claim**. This should be the answer to a well-formed question. When possible, your claim should include numbers.

Example of a bad claim:

As the temperature of the heat reservoir changed, so did the temperature of the metal cylinder attached to it.

Example of a better claim:

We measure a heat transfer coefficient of $(10.3 \pm 0.4) \text{ W}/(\text{m}^2 \cdot \text{K})$ between the heat reservoir and the metal cylinder.

- ☐ **Revise your figures** based on your claim. Do the figures present the right supporting information? Are all the figures relevant to the narrative?
- ☐ Think about your **target audience**. Consider the level of knowledge of the people you are writing for.

3 Steps to write a report or article

Reports and articles do not have to be written beginning to end. In fact, writing an introduction or an abstract should be impossible until many other components are present in the text. While there is no “one-size-fits-all” method, consider the steps below as a good starting point:

1. Prepare the figures and data tables. Upload them first into your report. Create informative captions for each of them.
2. Write the experimental/computational methods section.
3. Write your results and analysis. These can be in different sections or joined together.
4. Write your conclusion.
5. Write your theory section. Based on your analysis, figure out what math and physics you need to cover to bring your target audience to an understanding of the claim. You don’t have to spell everything out. You can refer other texts for lengthier derivations.
6. Write your introduction. Now that you have the full story, you can create an introduction that offers a starting point for your narrative.
7. Write the abstract.

No matter the order in which you write the report, you will have to iterate through the writing process. You should never consider a section “done” until the whole report is written and is coherent.

You may also consider creating an outline of your manuscript before you start writing (either the whole report or for individual parts).

4 What to write in a report or article

In this section, tips, tricks, and examples are given for each section of a report:

1. [Abstract](#)
2. [Introduction](#)
3. [Theory Section](#)
4. [Methods](#)
5. [Data and Analysis](#)
6. [Conclusion](#)
7. [Acknowledgments](#)
8. [References](#)

4.1 Abstract

An abstract should contain the following information:

- The **objective** of report
- The **method**
- The **result(s)** including an uncertainty whenever possible.
- A short **discussion** of the meaning of the result
- Possible **future direction**

Below is an example abstract:

*We demonstrate a high energy throughput, modular optical laser pulse shaping technique for generating tunable, narrowband, terahertz radiation from the surface of InAs. We achieve a frequency selectivity ($\Delta f/f$) of 0.10 at 1.18 THz and demonstrate an energy throughput of up to 98% using two etalons to create a sequence of optical pulses. In contrast with previously reported schemes, our technique does not rely on interferometry or involve diffractive optical elements, making it robust and relatively inexpensive to implement. **This technique can be expanded with additional etalons in order to achieve greater frequency selectivity without sacrificing efficiency.** Finally, we analyze the effects of carrier saturation within the InAs as it contributes to the spectral shape of the terahertz radiation and selectivity of this method.*

This abstract is missing a statement of uncertainty as this particular claim in this particular sub-field of physics does not require one. This is an exception.

4.2 Introduction

4.2.1 What to include

The introduction must answer the following questions:

- What **question** are you addressing?
- What is the **broader context** for the work presented? What has been done before?
- What is the **motivation** for the work presented?

In addition to answering those questions, the introduction should include:

- A clear **statement of what work was done**.
- A brief description of the **method**.
- Your results. What is your main **claim**?

Below is an abbreviated introduction as an example. (It is longer than what needs to appear in a report.)

The last 20 years have seen the development of increasingly more efficient terahertz (THz) generation [CITATIONS] and detection [CITATIONS] methods, which have been at the core of many new research avenues and technologies. Many physical processes in materials science exhibit sensitivity to THz-frequency radiation, or occur on timescales comparable to that of a THz pulse. Applications of THz technology require efficient and effective tools for engineering bright THz sources with customizable characteristics. Recent work aimed at applying THz technology to security [CITATIONS] and industrial [CITATIONS] applications has underscored the need for broadband sources. Other applications, however, such as coherent control of vibrational modes in molecular samples [CITATIONS], require access to narrowband, tunable THz sources.

Motivated by the extensive availability of optical components and alignment techniques in the optical regime, most methods for producing THz radiation with the desired temporal and spectral characteristics demonstrated to date involve control of the generation mechanism rather than shaping of the THz radiation itself. Manipulation of both the spatial [CITATIONS] and temporal characteristics of the optical laser pulse that drives the THz generation allows control of the resulting THz radiation. For example, narrowband, tunable sources of THz can be achieved by purely temporal shaping of the optical laser to create a pulse train [CITATIONS]. This optical pulse train creates a THz pulse train with a narrow spectrum whose width and center frequency can be controlled by adjusting the number of and timing between pulses in the train, respectively.

Several methods for creating customizable optical pulse trains have been applied to shaping THz. One such method is ...

Various THz generation methods have been explored using these pulse shaping for manipulating THz radiation. To date, efforts have concentrated on photo-conductive antennas [CITATIONS], air plasma [CITATIONS], semiconductor heterostructures [CITATIONS], and nonlinear optical processes [CITATIONS] as tunable THz sources... To the best of our knowledge, none of the work done on THz spectral shaping has used un-biased semiconductor chips as their emitter. Such THz emitters have a number of advantages: they can be used with non-amplified laser sources, they are

relatively low-cost, and they lack the need of photo-conductive antennas for supporting electronics, further simplifying the setup.

In this paper, we demonstrate a simple, non-interferometric, modular scheme for generating tunable, narrowband THz radiation using a highly efficient optical pulse shaping technique. We use a series of staggered, adjustable etalons to create an optical laser pulse train and focus it on the surface of an un-biased InAs chip to generate THz. Using a series of etalons permits a modular approach that allows the desired degree of narrowing of the spectrum, as well as relative suppression of unwanted spectral features, at the cost of additional optics. Using this approach, we demonstrate a throughput possibly as high as 98% with a bandwidth selectivity $\Delta f/f = 0.10$ at 1.18 THz. For comparison, we also test a similar scheme involving a single etalon followed by a Michelson interferometer... Finally, we study the contribution of carrier saturation effects in InAs, and how this affects the shaping capabilities of this modular approach.

4.2.2 Additional concerns

- While you must introduce your main claim in the introduction, do not mix those with your analysis and conclusion.
- The introduction should start with a more global point of view and zoom in on the particulars of your work.
- You can give a roadmap of your report at the end of the introduction.

4.3 Theory section

Depending on the experiment, the theory section can significantly vary in length. In pure experimental demonstration, it can be almost absent. In reports with a more extensive analysis, the theory section should include the information needed to explain the claims, using the data as proof.

As mentioned above, keep your audience in mind as you write this section.

The theory section **may include**:

- physical concepts that help explain the results. For example, a report about the magnetic properties of nickel should include an brief introduction to ferromagnetism.
- mathematical concepts important for the analysis. For example, if you use Fourier analysis to process your results, a quick description of what Fourier analysis is and does must be included.
- mathematical derivations important for the analysis. If your analysis involves deriving a specific value from the a set of measurements related by an equation, that equation must be included and explained. Some of the steps of the derivation must be shown.

The theory section **should not include**:

- data or analysis.
- information unrelated to the analysis. It is easy to put too much information in the theory section. To avoid that, you may use citations to include more in-depth treatment of certain topics. For example, if you Fourier transforms in your analysis, you should not include all of the Fourier transforms properties in the report. You should only highlight the one(s) you use directly.

4.4 Methods

4.4.1 Experimental methods

An experimental description should:

- include a least one schematic of the experiment. Photos of the experimental setup are usually not appropriate.
- include a **direct reference** to the figure in the text.
- include **labels** in the experimental schematic that are referenced in the caption and/or text.
- include a **numbered caption** to the figure.
- **describe the experimental setup in the text.**
- **emphasis on the parts of the experiment that affect the results.** For example, in most cases, we don't need to know the make of the laser, only its capabilities.
- include important limitations or concerns of the apparatus.

Below is an example experimental description:

Figure 1 shows a schematic of the experimental setup. The THz generation is driven by a 50 fs pulse from a Ti:Sapphire laser oscillator with center wavelength 800 nm, pulse energy ~ 8 nJ, and repetition rate 80 MHz. A beam splitter sends 4% of the incident pulse energy to be used for electro-optic detection of the shaped THz. The remainder of the power goes through the pulse shaping setup before being focused on a InAs chip for THz generation.

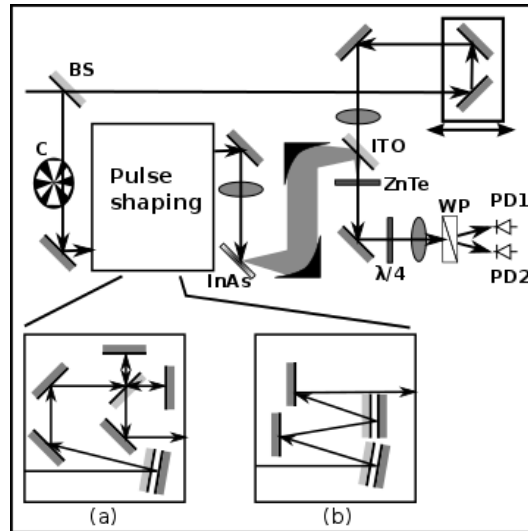


Figure 1: Schematic of the THz generation and detection setup. Two different pulse shaping techniques for the optical driving pulse are compared: (a) an etalon followed by a Michelson interferometer and (b) two etalons. **BS**: beam splitter. **C**: optical chopper. **InAs**: Indium Arsenide semiconductor chip. **ITO**: Indium Tin Oxide coated glass substrate. **ZnTe**: Zinc Telluride detection crystal. $\lambda/4$: quarter waveplate. **WP**: Wollaston prism. **PD1** and **PD2**: Photodiodes used for balanced detection.

Figure 1(a) shows the first pulse train generation scheme tested, an etalon followed by a Michelson interferometer. The etalon consists of a broadband dielectric partial reflector (39% reflective at normal incidence) and a high reflecting silver mirror...

Despite this overlap of pulses, we do not expect effects from interference between the two copies of the pulse train. The fourth pulse in the train has, at most, 10% of the intensity of the first pulse of the copied train. More importantly, however, in order to interfere, the two optical pulses must be overlapped in time and space to a degree of precision higher than required for this pulse train to effectively shape the THz. This is a result of the difference in time duration between the optical pulses (~ 50 fs) and the THz pulses (~ 1 ps). In fact, an advantage of our pulse shaping method is that we do not require the alignment precision and stability required in a interferometric scheme...

The THz radiation is steered and focused, using two off-axis, parabolic, gold-coated mirrors, onto a 1-mm thick ZnTe crystal for electro-optic detection [citation]. All measurements were performed in a nitrogen-purged environment with 3%-4% humidity to minimize absorption by water.

A common pitfall when writing the experimental description is to create a chronological list of the steps that were taken during the experiment. Avoid this. Describe the setup and the experiment, not your journey through it.

Schematic drawing tool: You must use software to make the experimental figure. There are two kind of drawing tools: pixel drawing software and vector drawing software. Vector graphic software is usually a little harder to learn but produces images that can scale without running into pixelation issues. Here is list of possible tools:

- Vector graphics: Inkscape (our recommendation), RollApp, LibreOffice Draw, Adobe Illustrator
- Pixel graphics: Krita, LibreOffice Draw, GIMP, Photoshop, PowerPoint
- Circuit diagrams: circuitikz, www.circuit-diagram.org, www.circuitlab.com

4.4.2 Computational methods

The methods section will look significantly different for an article reporting computational work. In such a paper, the methods section should include:

- The math involved to transition the information in the theory section into a model. For example, if you are solving a differential equation, you should write down how you simplified the equations and solved the derivatives in the methods section.
- Any approximation you made to simplify the equations and make them easier or more efficient to solve.
- The limits of the code in terms of resolution and/or stability.
- How long it takes to run the code under what conditions.
- A description of what specific simulation was run. Consider the example of simulating free fall. What were the initial velocity and position for each simulation?

4.5 Data and analysis

4.5.1 What to include

In this section, you use your data to prove your claims. You build your argument using the ideas you've developed in the theory and experimental description sections. You are telling a story of sorts. Think in terms of a narrative, not a step-by-step description of the math you did.

This section will include:

- **properly formatted figures and tables.** The format depends on the data being presented. Small data sets without a functional dependence can be presented in a table. Long data sets, or data sets where we expect a trend, should be presented in a graph.
- a **discussion of each graph in the text.** If you find that you have nothing to say about a figure, it might be a indication that it should not be there.
- a discussion of the **uncertainty in the results.** This discussion should include the likely sources of uncertainty and their effects on the data.
- a **comparison** between your results and those expected from primary sources when those comparisons are available.
- a discussion of your results and what they mean.

Below is an example:

Figure 2 shows the detected THz spectra obtained by using our two different pulse shaping configurations. In figure 2(a), narrow spectral peaks were observed when using the etalon-Michelson interferometer method for pulse shaping (see figure 1(a)). The position of the peak is determined by changing the spacing between the partial and high reflector in the etalon and the relative lengths of the arms of Michelson interferometer. The width of the peak is controlled by the number of pulses in the optical train. For comparison, the plot includes the broad THz spectrum measured when a single 800 nm pulse is incident on the InAs. The relative amplitude of the broadband spectrum to that of the narrowband peaks has been amplified for easier comparison. In figure 2(b), the 800 nm laser pulse was shaped using two etalons (see figure 1(b)). The normalized broadband spectrum from a single driving pulse is also shown for reference. The differences in shape of the broad spectra shown in figure 2 are typical of day-to-day variations in humidity and alignment.

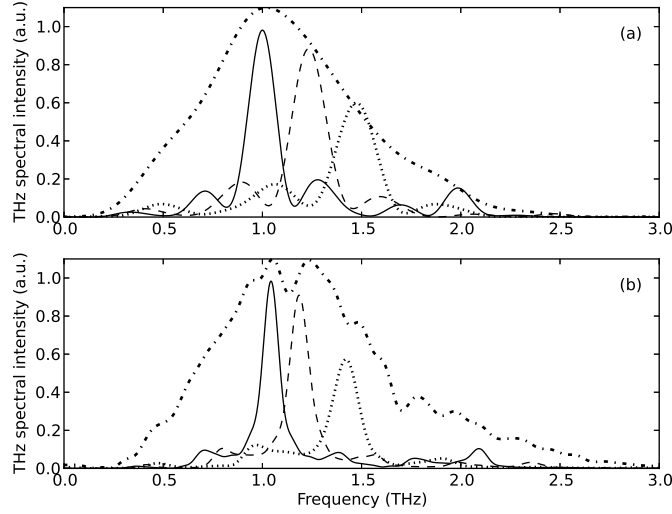


Figure 2: Data showing frequency selection for (a) the etalon followed by a Michelson interferometer pulse shaping configuration and (b) two consecutive etalons. The peaks displayed in (a) are centered at 0.99 THz, 1.22 THz, and 1.47 THz. The peaks in (b) are centered at 1.03 THz, 1.18 THz, and 1.41 THz. For both figures, the broad spectrum of the THz pulse arising from a single shaped 800 nm laser pulse is also shown.

Comparing the two sets of data in figure 2 clearly shows that we obtain better frequency selectivity in the case of two etalons compared to one etalon and one Michelson interferometer. The peak centered at 1.22 THz in figure 2(a) has $\Delta f/f = 0.16$ whereas the peak centered at 1.18 THz in figure 2(b) has $\Delta f/f = 0.10$. This can be readily explained... Saturation effects in the InAs, however, limit the use of simple Fourier transforms to accurately model the expected THz spectrum as is discussed later in the text.

The high energy throughput and mechanical stability of the two-etalon setup is quite favorable compared to other pulse shaping schemes. We compare our results to those of Vidal *et al.* [CITATIONS] (SLM in the frequency dispersed plane of a zero dispersion stretcher) and Chen *et al.* [CITATIONS] (etalon after a stretcher) as they are the most efficient methods to date. In the first case, ...

Another advantage of using only etalons for pulse shaping is the ease with which the technique can be expanded. A pulse shaping scheme using 3 or 4 etalons, for example,...

4.5.2 Additional concerns

Data analysis and visualization software: There are many types of software available to analyze and draw data. The list below splits the options into Open source and closed source categories. Open source options are usually free and equally powerful as the closed source versions but do not have the same polish in the user interface.

- Open source: Python is our recommendation for plotting, fits, filtering, and most other aspects of your analysis. Other resources include SciDAVis, QtiPlot, and Octave.
- Closed source: Origin, Matlab, Mathematica, Igor Pro, etc.

4.6 Conclusion

Your conclusion should:

- restate the results of the experiment.
- not present any new information.
- suggest future work.
- end with a broader view of your work and its significance.

Below is an example:

In conclusion, we have demonstrated the use of two staggered etalons as a high through-put pulse shaping method for THz generation. We used an un-biased InAs chip as our THz emitter and demonstrated bandwidth selectivity $\Delta f / f = 0.10$ at 1.18 THz. Carrier saturation effects in the InAs were shown to add additional pulse shaping to the THz electric field but not to prevent high selectivity in bandwidth. Our data indicate that the system could easily be expanded to include more etalons which would result in better frequency selectivity while maintaining a high energy efficiency. While useful for any laser system, this method is especially well suited for low-power laser systems that cannot afford the power loss associated with many other pulse shaping techniques and for researchers developing applications needing a modular approach.

4.7 Acknowledgments

Not all reports need an acknowledgement section, but it is a standard section for scientific articles wherein the authors acknowledge support from various sources, whether from a genuine sense of gratitude or by contractual obligation. For example, for a scientific study receiving funding from a particular government funding agency, private foundation, or commercial partner, the sources of funding must always be cited.

4.7.1 Informal support from people

In many cases, it is also a good idea to mention people who helped with the project but who did not meet the criteria to be included as an official coauthor on the paper. This could include people whom the authors asked for advice; people who gave feedback on drafts; and/or people who provided materials that enabled the work to occur. Of course, if a person contributed significant ideas and work to the report itself, they should generally be listed as a coauthor rather than merely acknowledged at the end. Different scientific journals and fields have different criteria for coauthorship: for example, the American Astronomical Society says:

All persons who have made significant contributions to a work intended for publication should be offered the opportunity to be listed as authors. This includes all those who have contributed significantly to the inception, design, execution, or interpretation of the research to be reported.

For your reports in this class, only the members of your group should be listed as “authors” of the report, but it may be appropriate to mention other classmates or people who provided ideas, suggestions, or feedback that you took into consideration when conducting your experiment or completing your report.

4.7.2 AI and LLMs

One essential element to include in your acknowledgements section is any use of artificial intelligence (AI) or large-language models (LLMs). These tools are often used in many different aspects of scientific work including the writing of analysis code, generation of schematic diagrams, as well as the generation, formatting, and editing of text. Most scientific journals have policies governing the acceptable use of AI in production of scientific studies.

For our course, using AI tools is allowed to help you make sure your analysis code is working properly, and it may also be used to help you with grammar and formatting. Any such use of AI should be described explicitly in the acknowledgement section of your paper (i.e., you should describe what parts of the analysis, writing, formatting, etc. you used these tools for and how you used them).

AI tools should *not* be used to generate text wholesale to be copied into your lab report: e.g., it should not be used to generate an introduction or a derivation that you include directly. Likewise, you should not use AI tools to generate code that you do not understand, as it may give erroneous results. At every stage, make sure that your submitted report reflects your own work and your own understanding of your results and their context.

4.8 References

References should appear at the end of the report and be formatted as:

- Names, *Title*, Publication, Date

Below are examples of *when* to use citations in your report.

In the introduction, use citations...

- to flesh out the broader context of your research.
- to highlight interest in the kind of research you are presenting.
- to point out past results relevant to your claim.

In the theory section, use citations to refer to long derivations you do not have room for in your report (where only a few of the steps should be shown) and background information that you assume the reader should be aware of but might still want to refresh on.

In the experimental description section, use citations...

- to show examples of similar experiment designs.
- to refer to more complicated experimental techniques that you employed in your work.

In the data and analysis section, use citations to compare your results to accepted values or similar measurements.

It is appropriate for a reference to be cited more than once in the report. For example, an article cited in the introduction about prior work done on your topic can be cited again in the analysis if you are comparing your results to those presented in the article.

5 Checklist: After you write

- ☐ Review every single sentence. For each sentence, determine whether or not it is needed. What is its purpose? Remove all unnecessary sentences.
- ☐ **Verb tense.** The work you did in the lab should be described in the past tense. Your results and conclusions should be written in the present tense since they are (hopefully) still valid.
- ☐ **Voice.** You should write in the active voice. For example, don't write:

The data were taken...

You can always tell that a sentence is written in the passive voice if you can add "... by vampires" at the end. Instead, write:

We used the apparatus to measure...

- ☐ Check your report for informal and/or imprecise language. Don't write:

We saw pretty good agreement between the theory and our experiment.

Instead, write:

The accepted value for the speed of sound in the lab is 343 m/s. We measured (335 ± 10) m/s.

Don't write:

We had to keep a careful eye on the temperature gauge to make sure things worked okay.

The assumption is that you did everything carefully. Instead, write:

We monitored the temperature during the experiment.

- ☐ Check the style guide.
- ☐ Put the paper down for at least half a day. Then re-read it with a fresh mind.
- ☐ If there is time, have another student read the report.
- ☐ If there is time, read the report to yourself slowly and out loud.

6 Style guides

When publishing a manuscript in a scientific journal, there are many rules to formatting text, figures, etc. These rules are covered in the style guides for the individual journals. In this course, we use the following rules:

6.1 Figures and Tables

- **All tables and figures must be mentioned explicitly by number** and appear in the correct numerical order. That is, Tables 1, 2, 3, and 4 must each be mentioned in the text at least once, and the first mention of Table 3 should not precede the first mention of Table 2.

Note: Don't directly type the number of any table or figure; have latex keep track of the numbers using `\label` and `\ref`. See examples in templates.

6.1.1 Figures

- **All figures must have a caption** and figure captions should clearly and concisely label and explain figures and parts of figures. The first sentence of each figure caption should be a descriptive phrase, omitting the initial article (the, a, an).
- **In multi-part figures, the captions should distinguish (a), (b), (c), etc., components of the figure.** Note that if parts are identified in the legend as (a), (b), (c), particularly for single figures composed of multiple panels, these letters should be clearly labeled in the figure itself. Otherwise panels should be referred to by position (top right, top left, middle, bottom, etc.)
- **All lines** (solid, dashed, dot-dashed, dash-dotted, etc.) **and symbols** (filled or open circles, squares, triangles, crosses, arrows, etc.) **should be explained in the caption.**

6.1.2 Tables

- **Every table must have a caption** and the caption should be a concise title (less than a sentence); more extensive descriptions or secondary information should be incorporated in a note to the table.
- **Each column, including the first, must have a heading.** Column headings should label the entries concisely (one or two words); the first letter of each word is capitalized.
- **Units of measurement should be given in parentheses immediately below the column headings**, not listed with the data in the body of the table.

6.2 Equations

- **Equations should not be referred to by their numbers alone**; that is, say “substituting in Eq. (45)” rather than “substituting in (45).”

Note: Don't directly type the number of any equation; have latex keep track of the numbers using `\label` and `\ref`.

- **Values given in scientific notation should be expressed with a multiplication sign** preceding the power of 10 (e.g., 3.4×10^{-18}); in tables only, to conserve space, the form 3.4E-18 may be used.
- **Stacked fractions are not permitted in the body of the text or in superscripts:** that is, inline and superscript fractions should be set as dt/ds , not $\frac{dt}{ds}$.

6.3 Miscellaneous

- **Acronyms and abbreviations should be spelled out the first time they are used** unless they are common throughout the discipline. Terms defined in the abstract should be defined independently in the main text.
- **Symbols for chemical elements are in normal type**, not italics. The mass number precedes the symbol (e.g., ^{12}C).
- **Use standard abbreviations for SI** (e.g., m, km, mm) **and natural units** (e.g., au, pc, cm). If English units such as inches or pounds per square inch are used, metric equivalents should follow in parentheses.
- **Avoid beginning sentences with a symbol, number, or lower-case letter.**
- **The word “data” is plural and takes a plural verb** (i.e. “these data are displayed in Fig. 2” rather than “this data is displayed in Fig. 2”).